

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0299

Nine Springs 138 kV
August 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0299. The Interconnection Customer proposes a 168.3 MW wind facility interconnecting at the Nine Springs 138 kV substation in the Maryland area. The request was received on December 31, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 174 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,968,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Rio Blanco 2702 Substation to Bruce Crossing Substation 138.0 kV is overloaded to 136.2% of its 235 MVA emergency rating for the loss of the Sierra Pacific Lumber 138/1kV Transformer. The project contributes 13.8% DFAX (23.2 MW) to this overload.

The Holopaw to Bruce Crossing Substation 138.0 kV is overloaded to 123.8% of its 276 MVA emergency rating for the loss of the Sierra Pacific Lumber 138/1 kV Transformer. The project contributes 13.8% DFAX (23.2 MW) to this overload.

Alliance to Rio Blanco 2702 138.0 kV L: 120.5% loading (246 MVA) following the outage of the Sierra Pacific Lumber 138/1 kV Transformer Bank; DFAX = 13.8%.

Westport to Burlington Ks 138.0 kV: 108.6% loading (255 MVA) following the outage of the Dubois Substation to Warren 138.0 kV L; DFAX = 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rio Blanco 2702 Substation to Bruce Crossing Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,098,000	100% (\$2,098,000)
Holopaw to Bruce Crossing Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$5,211,000	100% (\$5,211,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alliance to Rio Blanco 2702 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,217,000	100% (\$1,217,000)
Westport to Burlington Ks 138.0 kV	Reconductor line	\$442,000	100% (\$442,000)
	Total allocated to project		\$8,968,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.